

# PySpark Hands-on Practice (100 Questions)

## Dataset

```
df_emp (emp_id, name, dept_id, salary, manager_id, join_date)
df_dept (dept_id, dept_name)
df_customers (customer_id, name, city)
df_orders (order_id, customer_id, order_date, amount)
df_products (product_id, product_name, category, price)
df_sales (sale_id, product_id, quantity, sale_date)
```

## Questions

### BASIC (1-20)

1. Create SparkSession
2. Read CSV file into DataFrame
3. Show first 10 records
4. Print schema
5. Select specific columns
6. Rename a column
7. Add new column using withColumn()
8. Drop a column
9. Filter rows (salary > 50k)
10. Apply multiple filter conditions
11. Sort DataFrame by salary desc
12. Remove duplicate rows
13. Count total records
14. Get distinct values of a column
15. Handle NULL values
16. Replace NULL with default value
17. Change column data type
18. Convert string to date format
19. Create column using conditions
20. Write DataFrame to CSV/Parquet

### INTERMEDIATE (21-40)

21. Count employees per department
22. Find average salary per department
23. Get max salary per department
24. Get min salary per department
25. Calculate total revenue per customer
26. Group by multiple columns
27. Filter aggregated results
28. Count distinct values
29. Identify duplicate records
30. Apply multiple aggregations
31. Sort aggregated data
32. Find department with highest salary
33. Get first and last order date per customer
34. Calculate total sales per product
35. Group data by month
36. Find total orders per day
37. Customers with more than 2 orders
38. Calculate average order value
39. Find highest selling product
40. Department with highest average salary

### JOINS (41-60)

41. Inner join employees and departments
42. Left join customers and orders
43. Right join example
44. Full outer join
45. Self join (employee-manager)
46. Join three DataFrames
47. Find unmatched records
48. Anti join
49. Semi join
50. Join using multiple conditions
51. Broadcast join

52. Handle duplicate columns after join
53. Join on multiple keys
54. Filter after join
55. Aggregate after join
56. Optimize join for large datasets
57. Find employees without department
58. Find customers without orders
59. Join with window function
60. Combine join + aggregation

#### WINDOW FUNCTIONS (61-80)

61. Row number per department
62. Rank employees by salary
63. Dense rank
64. Top 3 employees per department
65. Running total of sales
66. Moving average (last 3 rows)
67. Lag function
68. Lead function
69. Difference between rows
70. First value per group
71. Last value per group
72. Partition and order
73. Multiple partitions
74. Identify duplicates
75. Consecutive records
76. Nth highest salary
77. Cumulative sum
78. Percentage contribution
79. Multiple window functions
80. Filter using window

#### REAL-TIME (81-100)

81. Customers who never ordered
82. Top 5 customers
83. Monthly revenue
84. Daily active users
85. Monthly active users
86. Repeat customers
87. Churn customers
88. Peak usage day
89. Fraud detection
90. Most sold product
91. Products never sold
92. Recommendation logic
93. Cart abandonment
94. Session duration
95. Duplicate transactions
96. Trending products
97. Month-over-month growth
98. High-value customers
99. Data skew handling
100. Optimize PySpark job